

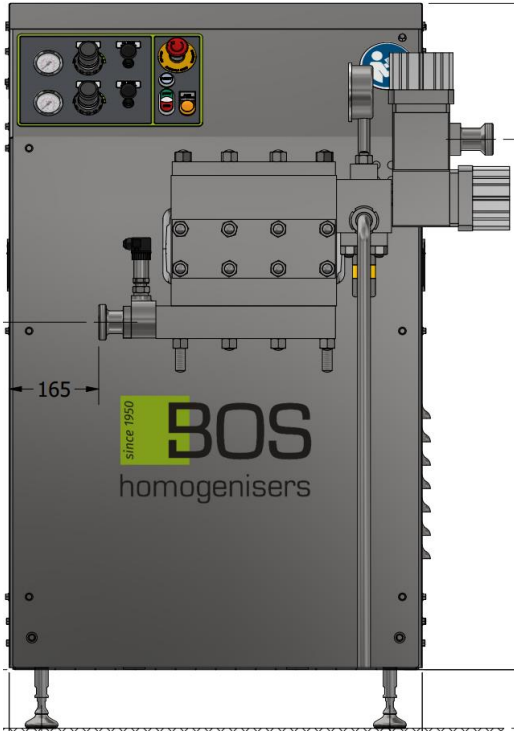
To:	Flow Force
Attention:	Umapathi
Date	21 September 2026
Ref.	-
Subject	H3-400P

We are pleased to quote:

One BOS Hilversum-Series High-pressure Homogeniser type H3-400P

Maximum capacity of this 3-piston homogeniser is 700 l/h at 400 bar, depending on installed motor.

The model has Poppet type pump valves and field replaceable seats made of hardened 17-4pH, suitable for moderately viscous and moderately abrasive products without any hard solids.

Machine type:	BOS high-pressure homogeniser H-Series H3-400P	
Product(s):	Water oil emulsion, no solids <i>Maximum particle size Low ($\leq 500 \mu\text{m}$)</i> <i>Fiber length Low (Fiber length $\leq 1,0\text{mm}$)</i>	
Capacity:	500 l/h	
Operating temperature:	90°C	
Operating pressure:	400 bar max. Total pressure	
Required feed pressure:	Min. 2 bar + 0,05 bar / % solids	
Plunger diameter:	25 mm – 3 piston	
Power consumption	6,5 kW	
Eccentric shaft speed	236 RPM	
		<i>Machine impression:</i>

Technical specifications:

Cylinder block:

- Forged “one-piece” cylinder block made of DUPLEX 1.4462 grade high tensile stainless steel. The design meets high sanitary standards, suitable for C.I.P.
- Ergonomic height for easy maintenance access, Compact machine design, small footprint.
- Poppet type, spring loaded, high efficiency pump valves with field replaceable seats, both valve and seat made of hardened 17-4ph / 1.4542 stainless steel alloy.
- Hardened stainless steel heat-treated 17-4ph plungers – 3 pieces (*Optional Chromium-carbide*)
- Plunger packing, spring loaded, suitable for <95°C, seals/gaskets NBR, UHMWPE, PEEK
- Plunger water cooling, integrated in the cylinder block, 1 L/min
- Analogue pressure gauge for Total pressure 0 – 400 bar (*Optional 4 – 20mA/digital*)
- Safety pressure relieve valve, mechanical/spring loaded, set at 120% of operating pressure.

Homogenising Valve:

- 2-stage Homogenising valve assembly, Pneumatic/Manual control. (*optional 2-stage*)
- Homogenising valve in 1st and 2nd stage made of hardened 17-4ph (*optional Chromium carbide*)
- Manual control of homogenising pressure, with local pneumatic valves in front panel. Analogue pressure gauges for pneumatic pressure, on/off buttons. (*Option Remote/Automatic control*)

Drive and sub-frame:

- Aluminum drive-base with integrated gear reduction, High mechanical efficiency, Low noise
- Splash lubrication of crankshaft, gear reduction and bearings, suitable for capacity variations of 50 – 100% under full load. Sight glass for lubrication oil level.
- Air cooled Drive and Lubrication oil, through natural convection/ventilation of motor compartment (no additional cooling water required)
- Stainless steel sub-frame design, hygienic feet (NGI®) 5-10 cm ground clearance
- Stainless steel machine cladding with removable panels. Compact design, small footprint.

Connections and requirements:

- Main motor: 11kW 400V-50Hz 3ph IE3 WEG®, incl. pulley drive for 500 l/h
- Pressurized air 6 – 8 bar , for homogenising valves
- Plunger cooling water 1,0 l/min incl. strainer
- Emergency stop
- Wiring of electronic components in the machine into a the Terminal box.
- *Optional Electrical Control panel with YD starter or VFD (basic or premium panel)*
- Product inlet DN32 DIN 11851
- Product outlet DN25 DIN 11851

Dimensions and weight:

- Height 125 cm. Depth 115 cm. Width 80 cm. Weight approx. 650 kg.

Delivery:

- Performance testing with water, and quality inspection before shipment. Test certificate.
- Machine will be supplied including lubrication oil, set of spare parts, instruction E-manual.
- Machine will be cleaned, shrink wrapped and packaging in Wooden Export crating.

Price for this BOS H-series Homogeniser:

~~€ 26.268,00~~

Discounted – 15%

€ 22.327,80

Optional Extra's:

Code:	Description:	Price:
HEC14	Main motor:	Included
HEC09	11kW up to 2200 liter per hour, 400V-50Hz	€ 313,-
HEC13	Alternative voltage	€ 569,-
	Prearrangement for capacity variation	€ 640,-
	High Efficiency IE4 main motor	
HCL01	Cylinder block made of DUPLEX high tensile stainless steel	Included
HCL18	Poppet type, spring loaded, high efficiency pump valves 17-4ph / 1.4542	Included
	Hardened stainless steel plungers – 3 pieces	Included
	Chromium-carbide coated plungers – 3 pieces , extra wear/corrosion resistance	€ 3.317,-
	FDA-compliant product contact gaskets/seals (excl. material certificates)	€ 995,-
HCL19	Aseptic Cylinder block, incl. double packed cylinder block, suitable for continuous aseptic production with Sterile water <95°C (customer supply)	€ 6.396,-
HCL10/11/12	Other Product inlet/outlet connections: SMS, Tri-clamp or DIN-118464-2 Aseptic	€ 370,-
HCL07/08	Stand Pipe with T-Piece DN40 DIN11851 for inlet and/or outlet (price per piece)	€ 938,-
	2-stage Pneumatic actuated homogenising valve assembly, manual/local control	Included
	Homogenising valve made of hardened 17-4ph	Included
HHV01	Homogenising valve 1 st stage made of Tungsten-carbide, extra wear resistant	€ 711,-
HPR01	Analogue Pressure gauge for Total pressure, sanitary membrane (0 – 400 bar)	Included
HPR02	Analogue Pressure gauge for second stage	€ 1.194,-
HPR03	Analogue Pressure gauge for outlet pressure	€ 1.194,-
HPR04	Pressure transducer for Inlet pressure (4 – 20mA)	€ 1.962,-
HPR05	Pressure gauge for Total pressure with 4 – 20mA output signal	€ 1.009,-
HPR06	Pressure transducer for 2 nd Stage with gauge (4 – 20mA)	€ 1.890,-
HPR07	Pressure transducer for Outlet pressure with gauge (4 – 20mA)	€ 1.890
	Calibration certificates (Per piece)	€ 313,-
HCL15	Safety pressure relieve valve, sanitary design, set at 120% of operating pressure	Included
	Extra pressure relieve valve for line pressure above 50 bar	€ 2.746,-
HAU06	Low oil level switch, Low oil pressure switch	Included
HAU07	Solenoid 24VDC for plunger cooling water activation, incl. Strainer	€ 441,-
HAU10	Flow-switch security for insufficient plunger lubrication/cooling water.	€ 441,-
HAU11	Air solenoid valve with switch (electrical panel needed)	€ 398,-
	Air solenoid valve	€ 441,-
HEC01	Manual-automatic control of the homogenizing pressure 1 st stage	€ 1.504,-
HEC02	Manual-automatic control of the homogenizing pressure 2 nd stage	€ 1.504,-
HEC03	ON/OFF control of the air pressure with solenoid valve for the 1st stage	€ 469,-
	Homogenizing valve	
HEC04	ON/OFF control of the air pressure with solenoid valve for the 2nd stage	€ 469,-

Extra options:

HEC15	Basic control panel + Machine wiring (cabinet mounted on the back of the machine) for fixed capacity Electricals: Siemens components, motor protection, alarms, labeled parts, wiring & component lists; IEC/CE compliant. Controls: Start/stop, reset, emergency stop, indicator lights (front-mounted). Pressure Control: Manual regulator + ON/OFF air switch for both stages. Signal Interface: Terminal block for hardwired RUN/STOP only. Feed Monitoring: Pre-wired for volt-free relay; system shuts off on low feed (signal provided by customer).	€ 3.539,-
HEC16	Electrical Panel - stainless steel IP55 cabinet Electricals: Siemens components, incl. star-delta starter (main motor), HMI: 4" Unified Basic Panel MTP400 (Linux-based), front-mounted Controller: Simatic S7-1200, Profinet option available. Connections: Terminal block for hardwired signals (Profinet not included). Remote I/O: ET200SP module (replaces hardwired drive terminals). Options: Automatic homogenizing pressure regulation with remote control. Exclusions: Cabling between panel and machine not included.	€ 5.387,-
HEC17	Electrical Panel prearranged for Variable Capacity - stainless steel IP55 cabinet Electricals: Siemens components, motor fuses, DOL starters (aux motors), full protection, hour meter, alarms, schematics & lists; IEC/CE compliant. HMI: 4" Unified Basic Panel MTP400 (Linux-based), front-mounted CPU: Simatic S7-1200 Remote I/O: ET200SP module (replaces hardwired drive terminals). Pressure Control: Manual/automatic regulation if option selected. Connections: Terminal block for direct hardwired signals (Profinet not included). Exclusions: No VFD; no cabling between panel, VFD, and machine.	€ 6.013,-
HEC18	PROFINET connection for signal exchange Signal exchange is performed via the Profinet protocol using a CP 1243-1 PN Communication Processor. IP address can be configured directly from the HMI.	€1.066,-
HEC19	IP55 Frequency Converter, Combined with Electrical Panel pre arranged for variable capacity. Loose execution. DANFOSS frequency converter in IP55 enclosure (included only with complete panel, not basic panel). Supplied loose for wall/MCC room installation by customer. Shielded cables between panel-VFD and VFD-machine not included.	€2.559,-
HEC20	Electrical cabinet mounted on the back of the machine.	€ 881,-
HEC23	Electrical panel for special voltage / frequency	€ 441,-
	Wooden Export crating for Machine	Included

General Terms & Conditions:

Prices:	Incl. Machine as specified on page 1-2
	Excl. Extra options, Transportation, Installation, Commissioning
Discount:	15% to dealer <i>(still needs to be deducted from the option list)</i>
Terms of Payment	30% with the order 70% before shipment
Delivery time:	12 - 14 weeks
Delivery terms:	EXW Hilversum, the Netherlands
Warranty period:	2 year with maximum of 8.000 operation hours.
Terms & Conditions:	Bos Homogenisers B.V. Supply Conditions July 2018 apply to this offer
Quotation validity:	30 days